DSC 206: Homework 2

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Assignment 1 *Let $P = \{p_1, \dots, p_n\} \subset \mathbb{R}$ with $p_1 \le p_2 \le \cdots \le p_n$.*

(1.a) Describe the optimal 1-center.

(1.b) Describe the optimal 1-mean.

(1.c) Give an example where the 1-median minimizer is not unique. Then, assuming n is odd, find $p^* = \arg\min_{p \in P} \sum_{i=1}^n |p_i - p|$ and prove your answer.

Solution. **(1.a) Optimal 1-center.** The 1-center cost is $f(x) = \max_i |p_i - x| = \max(x - p_1,\, p_n - x)$ since the points are sorted. This is minimized when both terms are equal, i.e.,

$$x^* = \frac{p_1 + p_n}{2}, \qquad \text{cost} = \frac{p_n - p_1}{2}.$$

(1.b) Optimal 1-mean. The 1-mean cost is $g(x) = \sum_{i=1}^n (p_i - x)^2$. Setting $g'(x) = -2\sum_{i=1}^n (p_i - x) = 0$ yields

$$x^* = \bar{p} = \frac{1}{n}\sum_{i=1}^n p_i.$$

Since $g''(x) = 2n > 0$, this is the unique minimizer.

(1.c) 1-median.

Non-uniqueness example. Take $n = 2$ with $P = \{0, 1\}$. Then $h(x) = |x| + |x - 1| = 1$ for every $x \in [0, 1]$, so every point in $[0, 1]$ is a minimizer.

Odd n. We will show $p^* = p_{(n+1)/2}$ is the median element of P.

Proof. The function $h(x) = \sum_{i=1}^n |p_i - x|$ is convex and piecewise linear. On the open interval (p_k, p_{k+1}) for $1 \le k \le n-1$ we have

$$h'(x) = \#\{i : p_i < x\} - \#\{i : p_i > x\} = k - (n - k) = 2k - n.$$

For odd n, it is negative for $k \le \frac{n-1}{2}$ and positive for $k \ge \frac{n+1}{2}$. Thus h strictly decreases on $(-\infty, p_{(n+1)/2}]$ and strictly increases on $[p_{(n+1)/2}, \infty)$. Hence the unique minimizer over all of

$\mathbb{R}$ is

$$p^* = p_{(n+1)/2}.$$

Assignment 2 *Let μ^* be the optimal k-center cost (with arbitrary centers in $\mathbb{R}^d$) and $\widehat{\mu}^*$ the optimal restricted k-center cost (centers must lie in P). Prove $\mu^* \le \widehat{\mu}^* \le 2\mu^*$.*

Solution. **Lower bound $\mu^* \le \widehat{\mu}^*$.** Every restricted clustering is also a feasible (unrestricted) k-clustering. Therefore the optimum over a smaller (restricted) feasible set is at least the optimum over the larger feasible set: $\mu^* = \min_{\text{all}} \text{cost} \le \min_{\text{restricted}} \text{cost} = \widehat{\mu}^*$.

Upper bound $\widehat{\mu}^* \le 2\mu^*$. Let $\{(c_i^*, C_i^*)\}_{i\in[1,k]}$ be an optimal unrestricted clustering with cost μ^*. For each i, choose any data point $\widehat{c}_i \in C_i^*$ as a restricted center. Keep the partition be the same. For any $p \in C_i^*$, the triangle inequality gives

$$d(p, \widehat{c}_i) \le d(p, c_i^*) + d(c_i^*, \widehat{c}_i) \le \mu^* + \mu^* = 2\mu^*,$$

where both bounds use the definition $d(q, c_i^*) \le \mu^*$ for all $q \in C_i^*$. Thus this restricted clustering has cost at most $2\mu^*$, and since $\widehat{\mu}^*$ is the minimum over all restricted clusterings,

$$\widehat{\mu}^* \le 2\mu^*.$$

Assignment 3 *Graph $G = (V, E)$ has three connected components on vertex sets $V_1 = \{v_1, \dots, v_{n/3}\}$, $V_2 = \{v_{n/3+1}, \dots, v_{2n/3}\}$, $V_3 = \{v_{2n/3+1}, \dots, v_n\}$. Provide three orthonormal eigenvectors of the Laplacian L for eigenvalue 0.*

Solution. For any connected component, the indicator vector of that component is a null vector of L, since $L\mathbf{1}_V = 0$ on each component (the row sums of L restricted to a component are zero). The three indicator vectors of V_1, V_2, V_3 have disjoint support, so they are mutually orthogonal. Normalizing each to unit length (each has $n/3$ ones) gives the orthonormal basis

$$\phi_i = \sqrt{\tfrac{3}{n}}\, \mathbf{1}_{V_i}, \qquad i = 1, 2, 3,$$

explicitly $(\phi_i)_j = \sqrt{3/n}$ if $v_j \in V_i$ and 0 otherwise. By construction $L\phi_i = 0$, $\|\phi_i\| = 1$, and $\langle \phi_i, \phi_j \rangle = 0$ for $i \ne j$.

Assignment 4 *Let S be a finite point set with centroid $\mu(S)$, and let T be disjoint from S. Show that*

$$\big\|\mu(S \cup T) - \mu(S)\big\| \le \frac{|T|}{|S| + |T|} \big\|\mu(T) - \mu(S)\big\|.$$

Solution. Write $|S| = m$ and $|T| = t$. By definition,

$$\mu(S \cup T) = \frac{1}{m+t} \sum_{x \in S \cup T} x = \frac{1}{m+t}\Big(\sum_{x\in S} x + \sum_{x \in T} x\Big) = \frac{m\,\mu(S) + t\,\mu(T)}{m+t}.$$

Therefore

$$\mu(S \cup T) - \mu(S) = \frac{m\,\mu(S) + t\,\mu(T)}{m+t} - \mu(S) = \frac{t\big(\mu(T) - \mu(S)\big)}{m+t},$$

and taking norms,

$$\big\|\mu(S \cup T) - \mu(S)\big\| = \frac{t}{m+t}\big\|\mu(T) - \mu(S)\big\| = \frac{|T|}{|S| + |T|}\big\|\mu(T) - \mu(S)\big\|.$$

The inequality follows.

Assignment 5 *For the weighted graph in the assignment:*

(5.a) Construct the graph Laplacian L.

(5.b) Compute the bipartition obtained by spectral clustering: give the second eigenvector and the induced partition.

Solution. **(5.a) Laplacian.** Reading the edge weights from the figure: heavy edges $(v_1, v_2) = 50$, $(v_1, v_3) = 50$, $(v_4, v_5) = 50$; light edges $(v_1, v_4) = 1$, $(v_1, v_5) = 1$, $(v_2, v_4) = 1$, $(v_2, v_5) = 1$, $(v_3, v_5) = 1$. The weighted degrees are

$$d_1 = 102,\ d_2 = 52,\ d_3 = 51,\ d_4 = 52,\ d_5 = 53.$$

Thus $L = D - W$ equals

$$L = \begin{pmatrix} 102 & -50 & -50 & -1 & -1 \\ -50 & 52 & 0 & -1 & -1 \\ -50 & 0 & 51 & 0 & -1 \\ -1 & -1 & 0 & 52 & -50 \\ -1 & -1 & -1 & -50 & 53 \end{pmatrix}.$$

(5.b) Spectral bipartition. The smallest eigenvalue of L is 0 with eigenvector $\mathbf{1}/\sqrt{5}$. The second-smallest eigenvalue vector ϕ_2 approximately separates the two clusters of strongly-connected vertices. Heavy edges of weight 50 form two tightly bound groups: $A = \{v_1, v_2, v_3\}$ joined through v_1, and $B = \{v_4, v_5\}$ joined directly. Light weight-1 edges only weakly connect A and B.

The Fiedler vector takes opposite signs on A and B. To leading order it is the indicator-difference vector orthogonal to $\mathbf{1}$. With $|A| = 3$, $|B| = 2$, requiring $\sum_i \phi_2(i) = 0$ and $\|\phi_2\| = 1$ gives

$$\phi_2 \approx \Big(\sqrt{\tfrac{2}{15}},\ \sqrt{\tfrac{2}{15}},\ \sqrt{\tfrac{2}{15}},\ -\sqrt{\tfrac{3}{10}},\ -\sqrt{\tfrac{3}{10}}\Big).$$

The induced bipartition by sign of ϕ_2 is

$$C_1 = \{v_1, v_2, v_3\}, \qquad C_2 = \{v_4, v_5\}.$$

This matches intuition: the spectral cut severs only weight-1 edges and preserves all weight-50 edges.

Assignment 6 ***Problem 6 (3 pts).*** *Compute the Jaccard similarities of each pair of* $\{1,2,3,4\}$, $\{2,3,5,7\}$, $\{2,4,6\}$.

Solution. Let $A = \{1,2,3,4\}$, $B = \{2,3,5,7\}$, $C = \{2,4,6\}$. Recall $J(X,Y) = |X \cap Y|/|X \cup Y|$.

$$J(A,B) = \frac{|\{2,3\}|}{|\{1,2,3,4,5,7\}|} = \frac{2}{6} = \frac{1}{3},$$
$$J(A,C) = \frac{|\{2,4\}|}{|\{1,2,3,4,6\}|} = \frac{2}{5},$$
$$J(B,C) = \frac{|\{2\}|}{|\{2,3,4,5,6,7\}|} = \frac{1}{6}.$$

Assignment 7 *Using HW figure, augment the signatures with* $h_3(x) = 2x + 1 \bmod 5$ *and* $h_4(x) = 3x - 1 \bmod 5$.

Solution. The hash values per row are

x	0	1	2	3	4
$h_3(x) = 2x + 1 \bmod 5$	1	3	0	2	4
$h_4(x) = 3x - 1 \bmod 5$	4	2	0	3	1

For each column S_j, the minhash with hash h is $\min\{h(r) : S_j(r) = 1\}$.

- S_1 has 1's at rows $\{0,3\}$: $h_3 = \min(1,2) = 1$, $h_4 = \min(4,3) = 3$.
- S_2 has 1's at rows $\{2\}$: $h_3 = 0$, $h_4 = 0$.
- S_3 has 1's at rows $\{1,3,4\}$: $h_3 = \min(3,2,4) = 2$, $h_4 = \min(2,3,1) = 1$.
- S_4 has 1's at rows $\{0,2,3\}$: $h_3 = \min(1,0,2) = 0$, $h_4 = \min(4,0,3) = 0$.

The full signature matrix is

	S_1	S_2	S_3	S_4
h_1	1	3	0	1
h_2	0	2	0	0
h_3	1	0	2	0
h_4	3	0	1	0

Assignment 8 *For the matrix in Figure 3 with* $h_1(x) = 2x + 1 \bmod 6$, $h_2(x) = 3x + 2 \bmod 6$, $h_3(x) = 5x + 2 \bmod 6$*:*

(a) Compute the minhash signatures.

(b) Which hashes are true permutations?

(c) Compare estimated and true Jaccard similarities for the six pairs.

Solution. **(a) Signatures.** The hash values are

x	0	1	2	3	4	5
$h_1 = 2x + 1 \bmod 6$	1	3	5	1	3	5
$h_2 = 3x + 2 \bmod 6$	2	5	2	5	2	5
$h_3 = 5x + 2 \bmod 6$	2	1	0	5	4	3

The columns have 1's at rows $S_1 = \{2\}$, $S_2 = \{0, 1\}$, $S_3 = \{2, 3, 4, 5\}$, $S_4 = \{0, 2, 5\}$. Taking min over those rows:

	S_1	S_2	S_3	S_4
h_1	5	1	1	1
h_2	2	2	2	2
h_3	0	1	0	0

(b) Permutations. h_1 takes values $\{1, 3, 5, 1, 3, 5\}$, hitting only odd residues; not a permutation. h_2 takes values $\{2, 5, 2, 5, 2, 5\}$; not a permutation. h_3 takes values $\{2, 1, 0, 5, 4, 3\}$, all six residues exactly once; h_3 *is* a true permutation. (We know $ax + b \bmod n$ is a permutation iff $\gcd(a, n) = 1$, and $\gcd(5, 6) = 1$ while $\gcd(2, 6) = \gcd(3, 6) > 1$.)

(c) Estimated vs. true Jaccards. True Jaccards from the columns:

$$J(S_1, S_2) = 0/3 = 0, \quad J(S_1, S_3) = 1/4 = 0.25, \quad J(S_1, S_4) = 1/3,$$
$$J(S_2, S_3) = 0/6 = 0, \quad J(S_2, S_4) = 1/4 = 0.25, \quad J(S_3, S_4) = 2/5 = 0.4.$$

Estimated Jaccards from the signature (fraction of hash rows on which the two columns agree):

$$\hat{J}(S_1, S_2) = 1/3 \; (h_2 \text{ only}),$$
$$\hat{J}(S_1, S_3) = 2/3; (h_2, h_3),$$
$$\hat{J}(S_1, S_4) = 2/3 \; (h_2, h_3),$$
$$\hat{J}(S_2, S_3) = 2/3 \; (h_1, h_2),$$
$$\hat{J}(S_2, S_4) = 2/3 \; (h_1, h_2),$$
$$\hat{J}(S_3, S_4) = 1.$$

Possible Reason: The estimates are systematically too high. Two reasons we can tell: (i) only h_3 is a true permutation; h_1, h_2 collapse rows into very few buckets ($\{1, 3, 5\}$ and $\{2, 5\}$

respectively), inflating the chance of accidental agreement and thus over-estimating similarity; (ii) the sample size of 3 hash functions is far too small to yield concentration around the true Jaccard. Using many independent true permutations would drive $\hat{J} \to J$ by the law of large numbers (each indicator $\mathbf{1}\{h(S_i) = h(S_j)\}$ is Bernoulli with mean exactly $J(S_i, S_j)$).